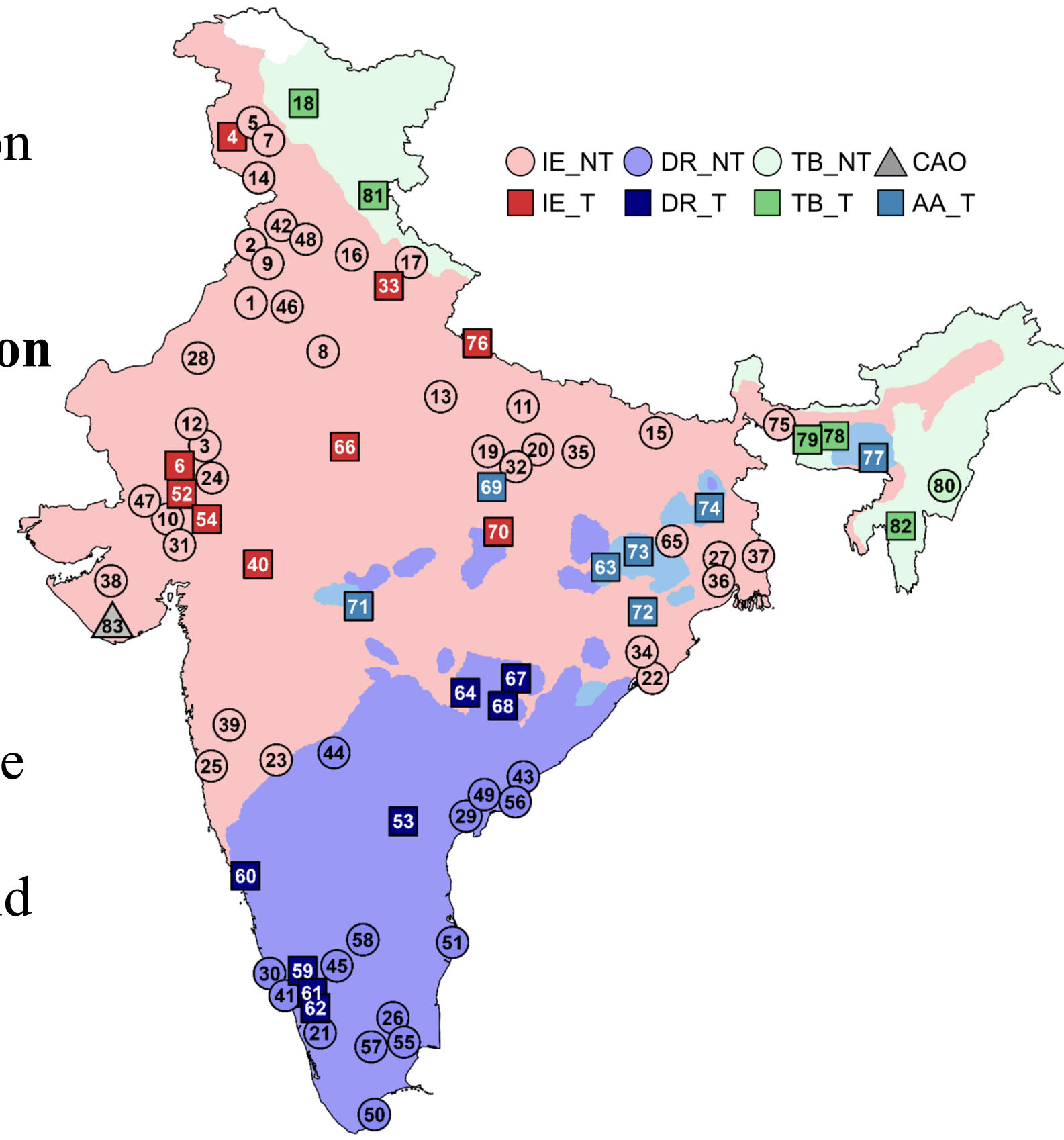


¹ Ankit Mukherjee, ² Chandrika Bhattacharyya, ^{1,3} Mohammed Faruq

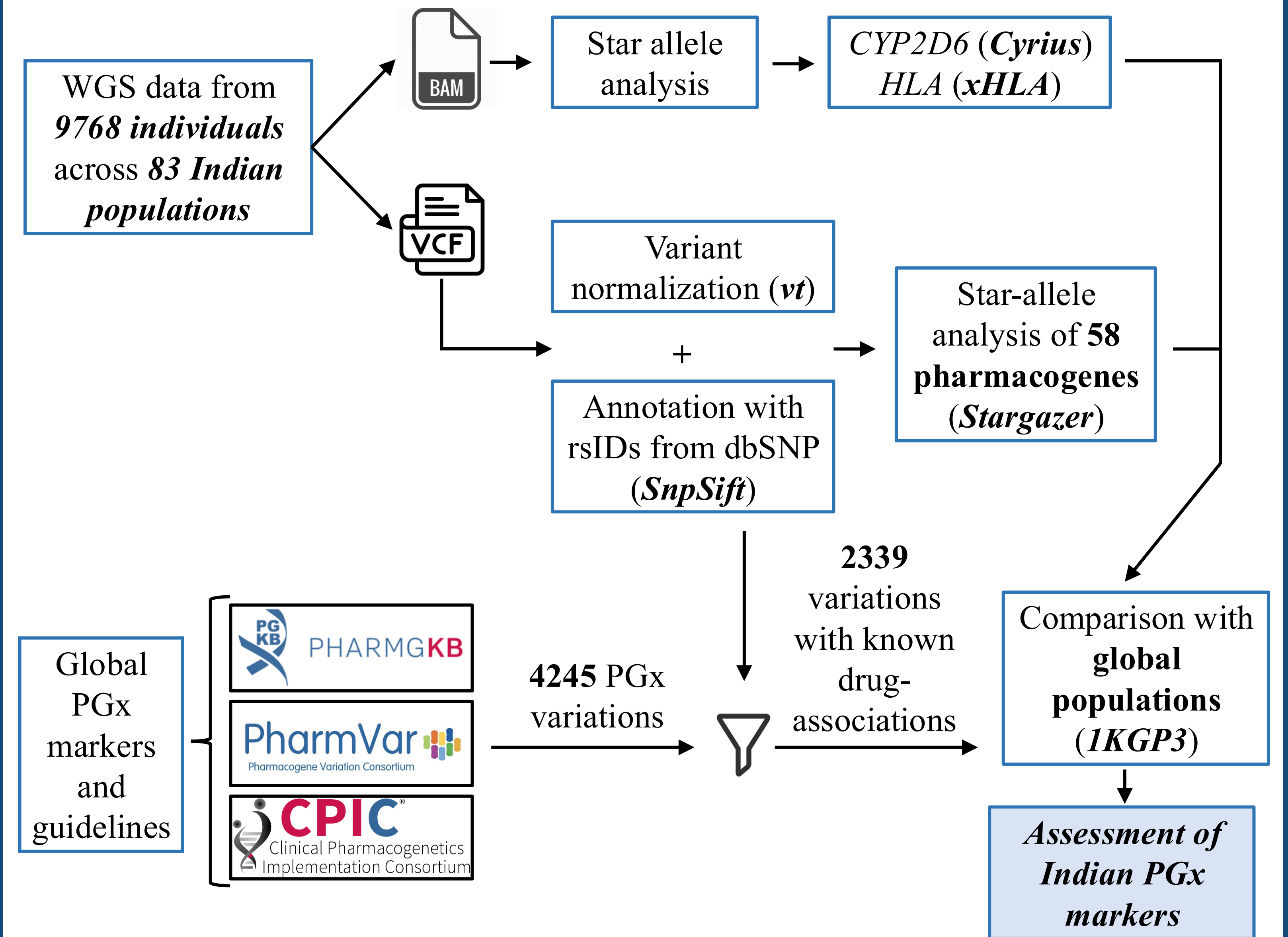
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BACKGROUND

- Pharmacogenomic variation influences **drug efficacy, toxicity** and **dosage** requirements while populations can differ substantially in the frequency and distribution of clinically relevant alleles.
- India presents a unique landscape for pharmacogenomic implementation because of its extensive genetic and ethnolinguistic diversity.
- Previous genomic resources, including **IndiGen** and **GenomeAsia**, have substantially expanded the characterization of pharmacogenomic variation in South Asian populations. However, **most analyses have provided limited resolution of variations across India's diverse populations.**
- The **GenomeIndia** project, comprising **9768 individuals from 83 populations**, provides an opportunity to characterize the population-level distribution of clinically actionable pharmacogenomic variation and evaluate its implications for precision medicine in India.

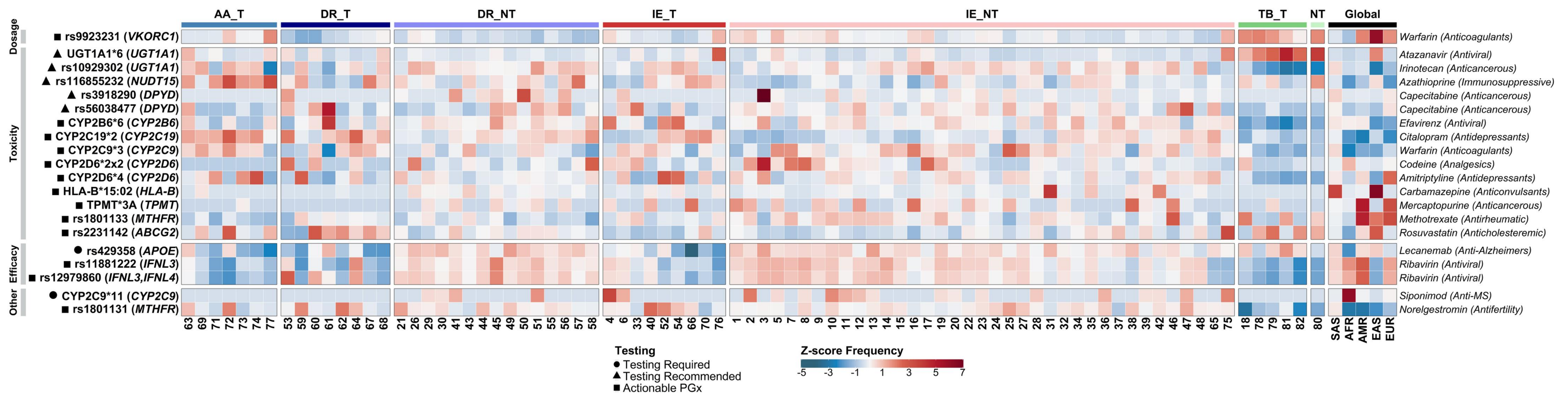


STUDY DESIGN

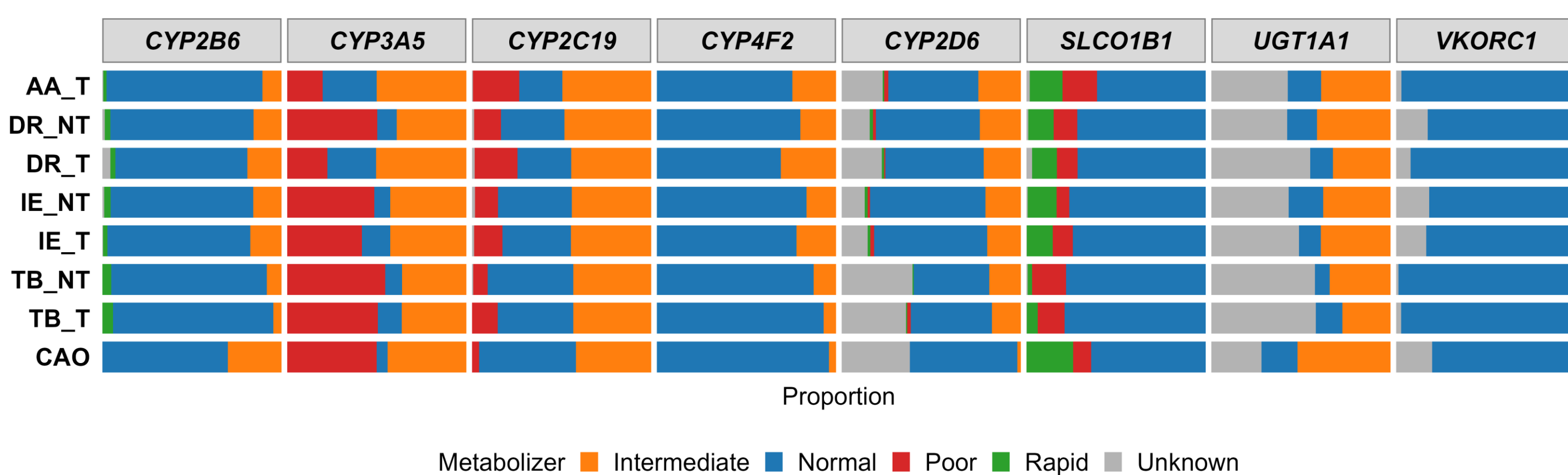


RESULTS

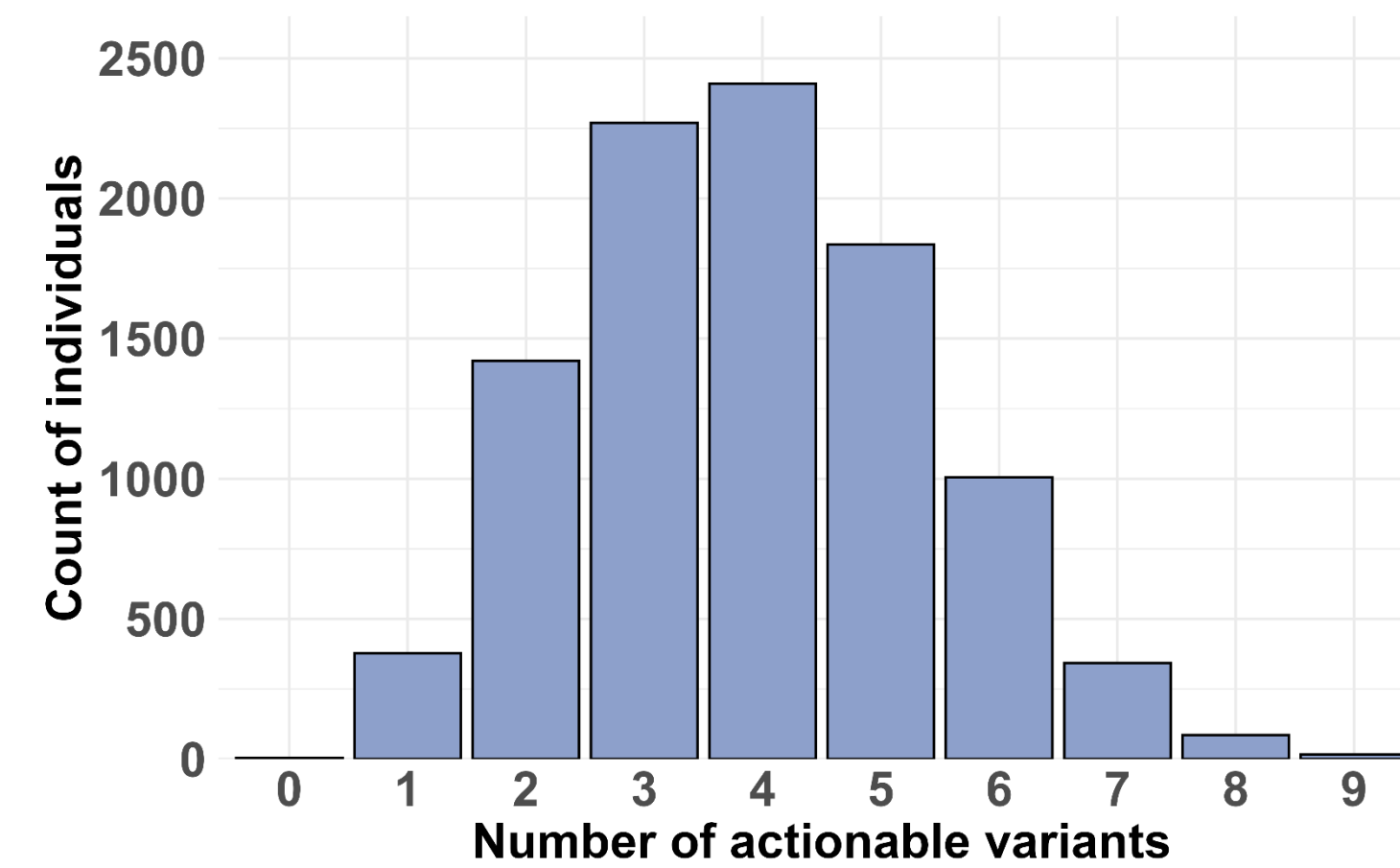
20 CLINICALLY ACTIONABLE PHARMACOGENOMIC VARIANTS



LINGUISTIC GROUP SPECIFIC METABOLIZER PHENOTYPES



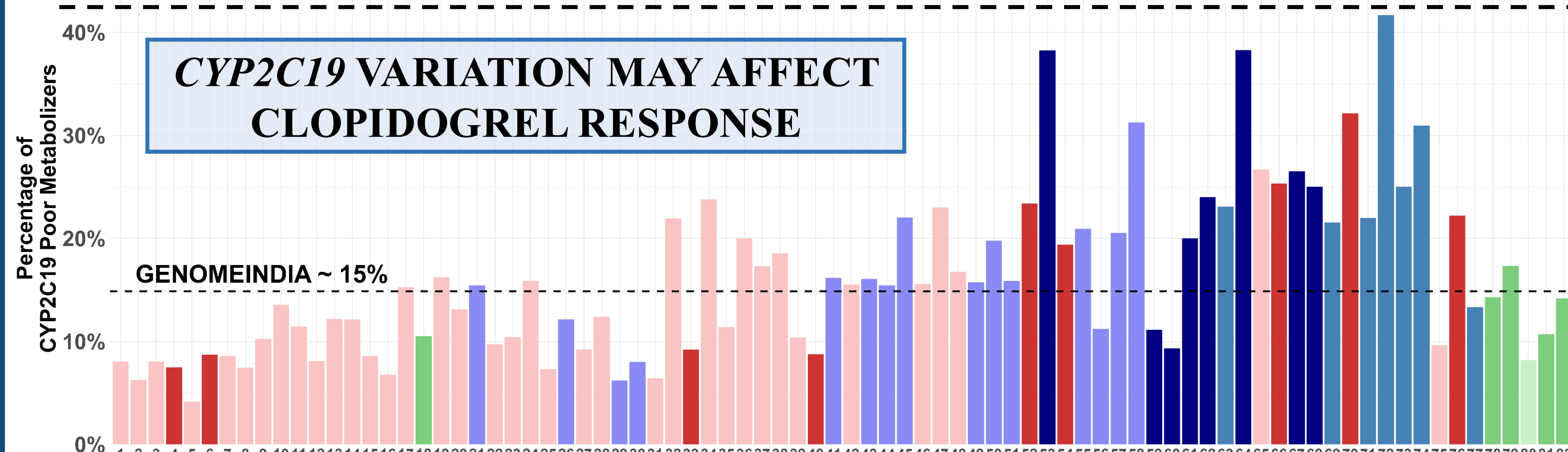
MEDIAN OF 4 ACTIONABLE PGx VARIANTS PER INDIVIDUAL



KEY THERAPEUTIC AREAS

Anticoagulants <i>VKORC1</i> (Warfarin)	Antineoplastics <i>DPYD</i> (Capecitabine)
Antivirals <i>UGT1A1</i> (Atazanavir)	Antidepressants <i>CYP2D6</i> (Amitriptyline)

CYP2C19 VARIATION MAY AFFECT CLOPIDOGREL RESPONSE



INCRETIN PATHWAY VARIANTS

Variant	Gene	Responder	% GI
rs10305420	GLP1R	Enhanced	31
rs6923761	GLP1R	Reduced	28
rs1800437	GIPR	Adverse	26

CYP2D6 STAR ALLELES IN GENOMEINDIA

Rare CYP2D6 haplotypes <i>*86, *111, *112, *113, *99</i>	Unknown Metabolizers % GenomeIndia = 18 Indigen = 9 SAS = 10
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CONCLUSION

GenomeIndia reveals substantial population-level heterogeneity in clinically actionable pharmacogenomic variation highlighting the need for population-aware PGx strategies

Key pharmacogenes such as *CYP2C19*, *CYP2D6*, *VKORC1* and *DPYD* warrant deeper population-specific investigation and clinical validation in India

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